

COMMON COURSE ASSIGNMENT

Cloud-Based College Event Registration System

A. Assignment Information

Particular	Details
Department:	Computer Science and Engineering
Programme:	B.Tech
Course Code & Course Name	Cloud Computing
Academic Year/Batch	2026-2027
Faculty Name	Dr. K. Santhiya Lakshmi
Assignment Title	Cloud-Based College Event Registration System
Date of Issue	02/09/2026
Date of Submission	02/09/2026
Maximum Marks	100
Course Outcome(s) – CO	CO1,CO2,CO3
Bloom's Taxonomy Level	L3 – Apply, L4 – Analyze, L5 – Evaluate
SDG Mapping (SDG 1-17)	SDG 4 – Quality Education; SDG 9 – Industry, Innovation and Infrastructure
Industry / Societal Relevance	Digital campus event management, cloud-based service delivery and student engagement

B. Assignment Problem / Challenge

Colleges conduct various technical, cultural, academic, sports and extracurricular events throughout the academic year. Managing student registrations manually using paper forms or separate spreadsheets can make it difficult to maintain centralized records, track participation, identify popular events and generate useful statistics.

The challenge is to design and develop a simple Cloud-Based College Event Registration System using Zoho Creator. The system should allow students to register for college events through an online form. The registration information should be stored and managed through the cloud.

The system should also provide administrators or event coordinators with reports showing the total number of registrations, event-wise registrations and department-wise participation. A simple dashboard/chart should be used to identify the most preferred or popular event.

The assignment requires students to apply cloud computing concepts, particularly Software as a Service (SaaS), cloud data storage, cloud application development, reporting and data visualization, to a practical college event-management problem.

C. Problem Statement

Problem/Case/Design Challenge

Design and develop a simple Cloud-Based College Event Registration System using Zoho Creator to demonstrate Software as a Service (SaaS).

The system will allow students to register for college events by providing their basic details, selecting an event and choosing their participation category.

The registration form shall collect the following information:

- Student Name
- Register Number
- Department
- Year
- Event Name

- Participation Type
- Contact Details
- Registration Date
- Registration Status

The submitted registration information shall be stored in the cloud and organized for easy retrieval and management.

An administrator or event coordinator shall be able to:

1. View registration records.
2. Search and filter registrations.
3. Update registration status.
4. View total registrations.
5. Generate event-wise registration reports.
6. Generate department-wise participation reports.
7. Identify the most popular event.
8. Display simple charts and dashboard information.

The proposed system demonstrates how a cloud platform such as Zoho Creator can be used to develop and deploy a simple SaaS-based college management application.

D. Requirements and Constraints

1. Functional Requirements

- Provide an online event registration form for students.
- Collect Student Name.
- Collect Register Number.
- Allow students to select their Department.
- Allow students to select their Year.

- Allow students to select an Event Name.
- Allow selection of Participation Type.
- Collect Contact Details.
- Automatically or manually record the Registration Date.
- Maintain Registration Status.
- Store submitted registration records in the cloud.
- Provide an All Registrations report.
- Provide an Event-wise Registration report.
- Provide a Department-wise Participation report.
- Display total number of registrations.
- Identify the most popular event.
- Provide a simple chart/dashboard for analysis.
- Allow administrators to filter registration records.

2. Non-Functional Requirements

Usability

The registration interface should be simple and easy for students to understand.

Reliability

Submitted registration information should be stored correctly and should be available when required.

Accessibility

The application should be accessible through internet-connected devices.

Security

Student and contact information should be accessible only to authorized users.

Scalability

The system should support an increasing number of student registrations and college events.

Maintainability

Administrators should be able to manage registration records and events without rebuilding the complete application.

3. Constraints

- The system depends on internet connectivity.
- The system depends on the availability of the Zoho Creator cloud platform.
- Some features may depend on the selected Zoho Creator plan.
- Student contact information must be protected from unauthorized access.
- The project should remain simple enough for academic demonstration.
- Sample data should be realistic and sufficient for testing reports and charts.
- Administrative access should be restricted to authorized users.

E. Student Work

1. Problem Understanding and Formulation

Problem

Manual event registration makes it difficult to maintain centralized student registration records, track participation, compare event popularity and generate accurate statistics.

For example, if students register through paper forms, the event coordinator may have to manually count registrations and organize students according to department and participation type.

Existing Process

Student → Paper/Manual Registration → Event Coordinator → Manual Record → Manual Counting

Proposed Process

Student → Online Registration Form → Zoho Creator Cloud → Cloud Registration Data → Reports/Dashboard → Event Coordinator

Expected Outcomes

1. Digitize college event registration.
2. Store registration information in the cloud.
3. Centralize student registration records.
4. Organize records according to event, department, year and participation type.
5. Allow administrators to track registration status.
6. Generate total registration reports.
7. Generate event-wise registration reports.
8. Generate department-wise participation reports.
9. Identify the most popular event.
10. Display simple charts for easier analysis.
11. Demonstrate practical use of SaaS.

Sample Data Fields

Field	Example
Student Name	Ananya
Register Number	23CSE101
Department	Computer Science
Year	II Year
Event Name	Hackathon
Participation Type	Participant
Contact Details	9876543210
Registration Date	02-09-2026
Registration Status	Confirmed

2. Application of Course Knowledge

Software as a Service (SaaS)

The application is accessed through the internet as a cloud-hosted service. Students and administrators can use the event registration application without installing and maintaining the complete software system on their local computers.

Zoho Creator provides the cloud-based environment required for creating and using the application.

Platform as a Service (PaaS)

Zoho Creator provides cloud application development tools that allow the creation of forms, reports, dashboards and application components without requiring the project team to manage the underlying physical infrastructure.

Cloud Accessibility

Students and authorized administrators can access the application through internet-connected devices. This makes the registration process more flexible compared with a paper-based system.

Cloud Data Management

Registration records are stored and managed through the cloud. Records can be organized using:

- Event Name
- Department
- Year
- Participation Type
- Registration Date
- Registration Status

This allows administrators to search, filter and analyze student participation.

Data Visualization

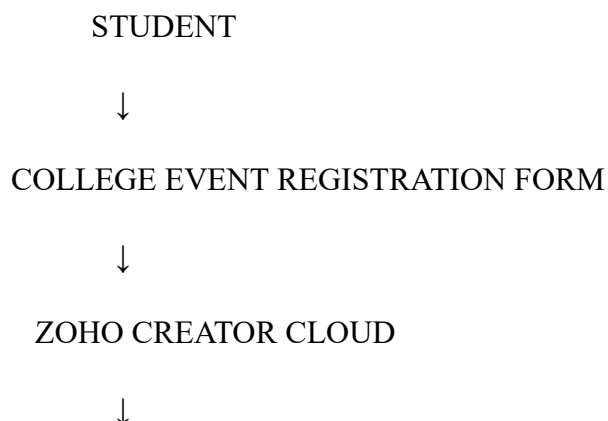
Charts and dashboard components can be used to represent registration information visually. This makes it easier to identify the most popular event and compare participation between departments.

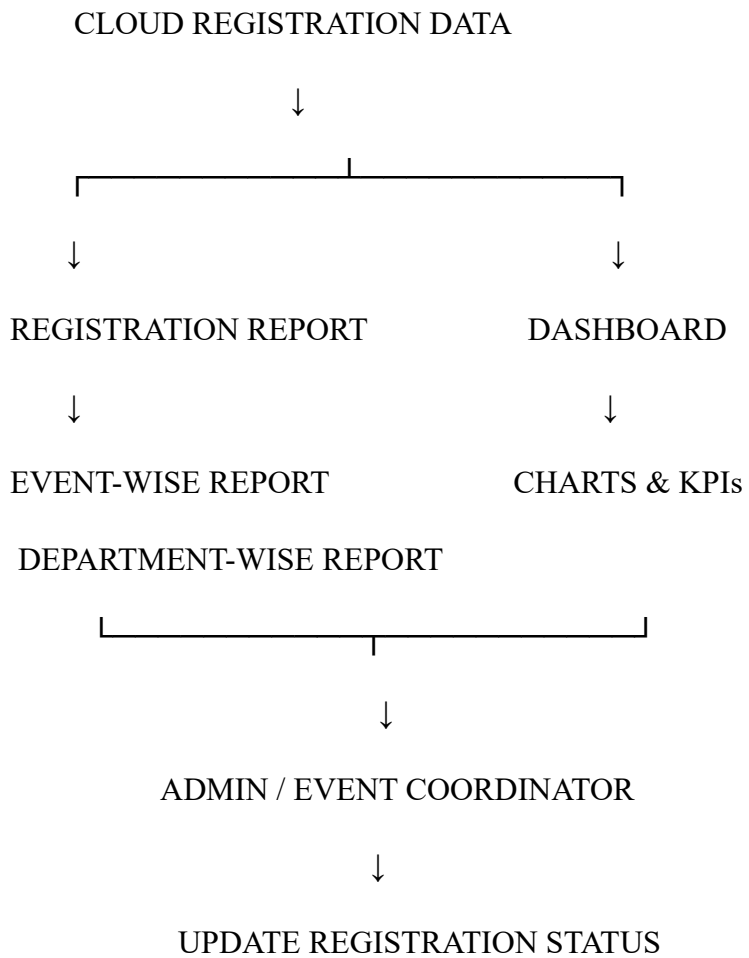
3. Solution / Design / Methodology

Major Modules

Module	Description
Module 1 – Student Registration	Students enter their details, select an event and participation type, and submit the registration form.
Module 2 – Cloud Registration Storage	Submitted registration information is stored in the Zoho Creator cloud application.
Module 3 – Registration Management	Administrators can view, search, filter and update registration records.
Module 4 – Reports and Dashboard	Reports display total registrations, event-wise registrations and department-wise participation.
Module 5 – Event Analysis	Charts are used to identify the most popular event and analyze student preferences.

System Architecture





Registration Status Workflow

New Registration



Registered



Confirmed



Event Participation

Registered



Cancelled

Reports Required

The system should generate the following reports:

1. Total Registrations

Displays the total number of students who have registered.

2. Event-wise Registrations

Groups students according to the event they selected.

Example:

Event Name	Number of Registrations
Hackathon	5
Cultural Fest	3
Tech Quiz	2
Sports Meet	1
Paper Presentation	1

3. Department-wise Participation

Displays the number of participants from each department.

Department	Number of Participants
CSE	5
ECE	3
AI&ML	3
IT	2

4. Most Popular Event

The event having the highest number of registrations is identified as the most popular event.

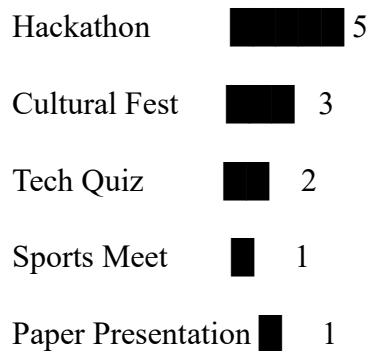
Most Popular Event: Hackathon

5. Simple Chart / Dashboard

A bar chart or pie chart can be used to represent event-wise registration.

Example:

Event-wise Registration



4. Use of Modern Tools

Zoho Creator will be used as the primary cloud application development platform.

The project will use the following capabilities:

- Create the College Event Registration Form.
- Create and manage registration fields.
- Store registration information in the cloud.
- Create an All Registrations report.
- Create Event-wise Registration report.
- Create Department-wise Participation report.
- Create dashboard cards.

- Create charts.
- Filter registration records.
- Update registration status.
- Manage administrative access.

Evidence of tool usage should be included through screenshots of:

1. Zoho Creator application.
2. Registration form.
3. Submitted registration records.
4. Registration report.
5. Event-wise report.
6. Department-wise report.
7. Dashboard/chart.

5. Results and Validation

The proposed system can be tested by entering representative student registration records and verifying whether the information is correctly stored in the cloud.

The generated reports can then be checked to determine:

- Total registrations.
- Number of registrations for each event.
- Department-wise participation.
- Participation type distribution.
- Registration status.
- Most popular event.

Sample Test Dataset

ID	Student	Department	Event	Participation	Status
R001	Ananya	CSE	Hackathon	Participant	Confirmed
R002	Rahul	ECE	Tech Quiz	Participant	Registered
R003	Priya	CSE	Cultural Fest	Participant	Confirmed
R004	Arjun	IT	Hackathon	Participant	Confirmed
R005	Meena	AI&ML	Paper Presentation	Presenter	Registered
R006	Vivek	ECE	Sports Meet	Participant	Confirmed
R007	Diya	CSE	Hackathon	Volunteer	Confirmed
R008	Kiran	IT	Tech Quiz	Participant	Registered
R009	Nisha	AI&ML	Cultural Fest	Participant	Confirmed
R010	Sanjay	CSE	Hackathon	Participant	Confirmed
R011	Asha	ECE	Cultural Fest	Participant	Registered
R012	Rohan	AI&ML	Hackathon	Participant	Confirmed

Dashboard Example

Category	Number
Total Registrations	12
Hackathon Registrations	5
Cultural Fest Registrations	3
Tech Quiz Registrations	2
Sports Meet Registrations	1

Category	Number
Paper Presentation Registrations	1
CSE Participation	5
ECE Participation	3
AI&ML Participation	3
IT Participation	2
Most Popular Event	Hackathon

Chart / Report Interpretation

Based on the sample dataset, Hackathon has the highest number of registrations with 5 participants. Cultural Fest has 3 registrations, Tech Quiz has 2 registrations, while Sports Meet and Paper Presentation have 1 registration each.

Therefore, the analysis identifies Hackathon as the most preferred event among the sample students.

Department-wise analysis shows that CSE has the highest participation in the sample data.

This information can help college event coordinators understand student preferences and plan resources accordingly.

Testing and Validation

Test Case	Expected Result
Submit valid registration	Registration should be stored successfully.
Enter Student Name	Student name should be stored correctly.
Enter Register Number	Register number should appear correctly in the record.
Select Department	Selected department should be recorded.

Test Case	Expected Result
Select Event	Selected event should appear correctly.
Select Participation Type	Participation type should be recorded.
Update Registered to Confirmed	Registration status should update correctly.
View Registration Report	Submitted registration should appear in the report.
Filter by Department	Relevant department registrations should be displayed.
View Event-wise Report	Registration count should be grouped by event.
Open Dashboard	Dashboard should display updated statistics.
Identify Popular Event	Event with the highest registration count should be identified.

6. Analysis and Engineering Decision

The proposed cloud-based event registration system is preferable to a manual registration system because it centralizes student registration information and provides faster tracking, filtering and analysis.

Feature	Manual System	Cloud-Based System
Registration	Paper/manual	Online
Data Storage	Registers/files	Cloud
Tracking	Difficult	Easy
Searching	Manual	Fast filtering/search
Statistics	Manual calculation	Dashboard
Event Comparison	Difficult	Event-wise report

Feature	Manual System	Cloud-Based System
Department Analysis	Manual	Department-wise report
Accessibility	Limited	Internet-based
Status Monitoring	Difficult	Centralized
Data Analysis	Time-consuming	Quick

Zoho Creator is selected because it provides a suitable cloud environment for creating forms, managing data, generating reports and creating dashboards.

It is therefore appropriate for demonstrating SaaS and cloud-based application development at an academic level.

7. Broader Considerations

Sustainability

Digital registration reduces the requirement for paper-based forms and manual documentation.

Environment

Reduced paper usage contributes to more environmentally friendly event administration.

Society

The system provides an organized communication and registration mechanism between students and college event coordinators.

Accessibility

Students can register for events through internet-connected devices without physically visiting an event office.

Security and Privacy

Student information and contact details should be protected from unauthorized access.

Economics

Centralized digital management can reduce manual administrative effort and paper-related costs.

Professional Responsibility

Registration records should be maintained accurately and should not be modified or disclosed without proper authorization.

8. Conclusion

The Cloud-Based College Event Registration System provides a practical approach to digitizing college event registration using cloud computing technology.

Students can register for events through an online form by entering their basic details, selecting an event and choosing their participation type. The registration information is stored and managed through the cloud using Zoho Creator.

The system provides total registration information, event-wise registration reports and department-wise participation reports. A dashboard and simple charts can be used to identify the most popular event and understand student participation patterns.

The project demonstrates the practical application of Software as a Service (SaaS), cloud storage, online application access, cloud-based application development, reporting and data visualization.

The proposed system can reduce manual record keeping, improve registration management and provide event coordinators with useful information for planning future college events.

9. Student Reflection

Through this assignment, I learned how cloud computing concepts can be applied to solve a real-world problem. I understood the working of the SaaS model and learned how a cloud-based application can collect, store and analyze information.

I also learned how forms, reports and dashboards can be used to create a simple management system using a cloud platform.

If additional time and resources were available, the system could be improved by adding automatic notifications, administrator roles, complaint escalation, image upload for complaints and student feedback after complaint resolution.

10. References

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3. Zoho Corporation. *Zoho Creator Documentation and Product Resources*.
4. Rittinghouse, J. W., & Ransome, J. F. (2017). *Cloud Computing: Implementation, Management, and Security*. CRC Press.
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